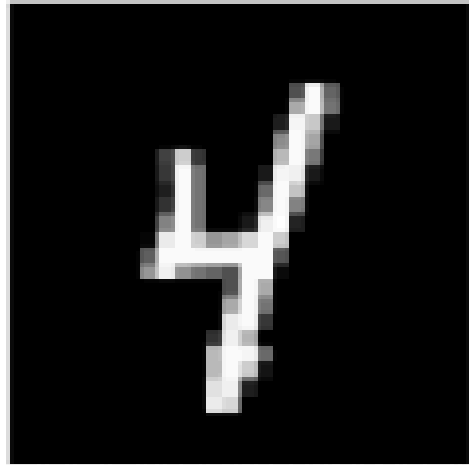
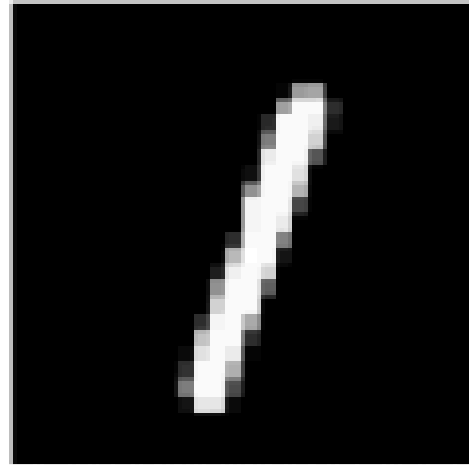


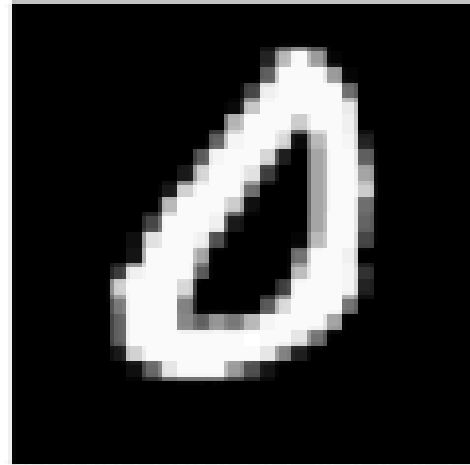
Les convolutions : intuition



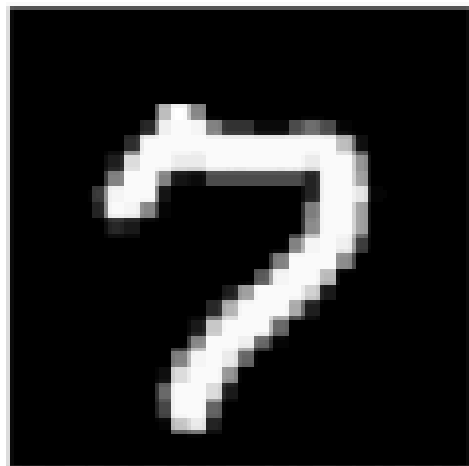
4 (4)



1 (1)



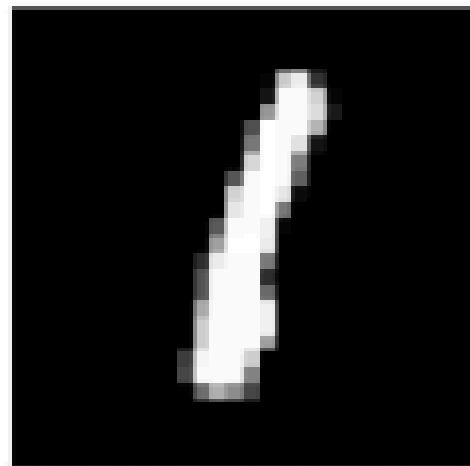
0 (0)



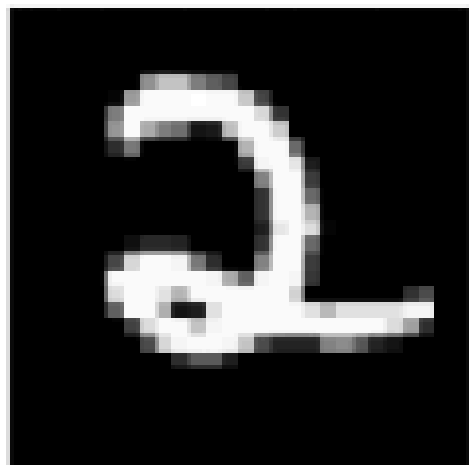
7 (7)



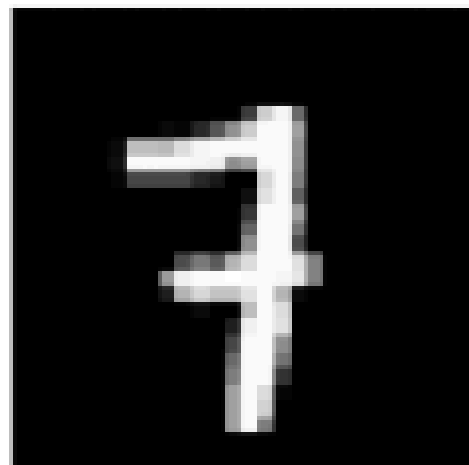
8 (8)



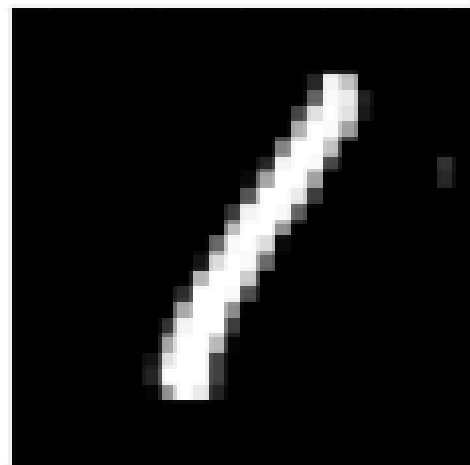
1 (1)



2 (2)



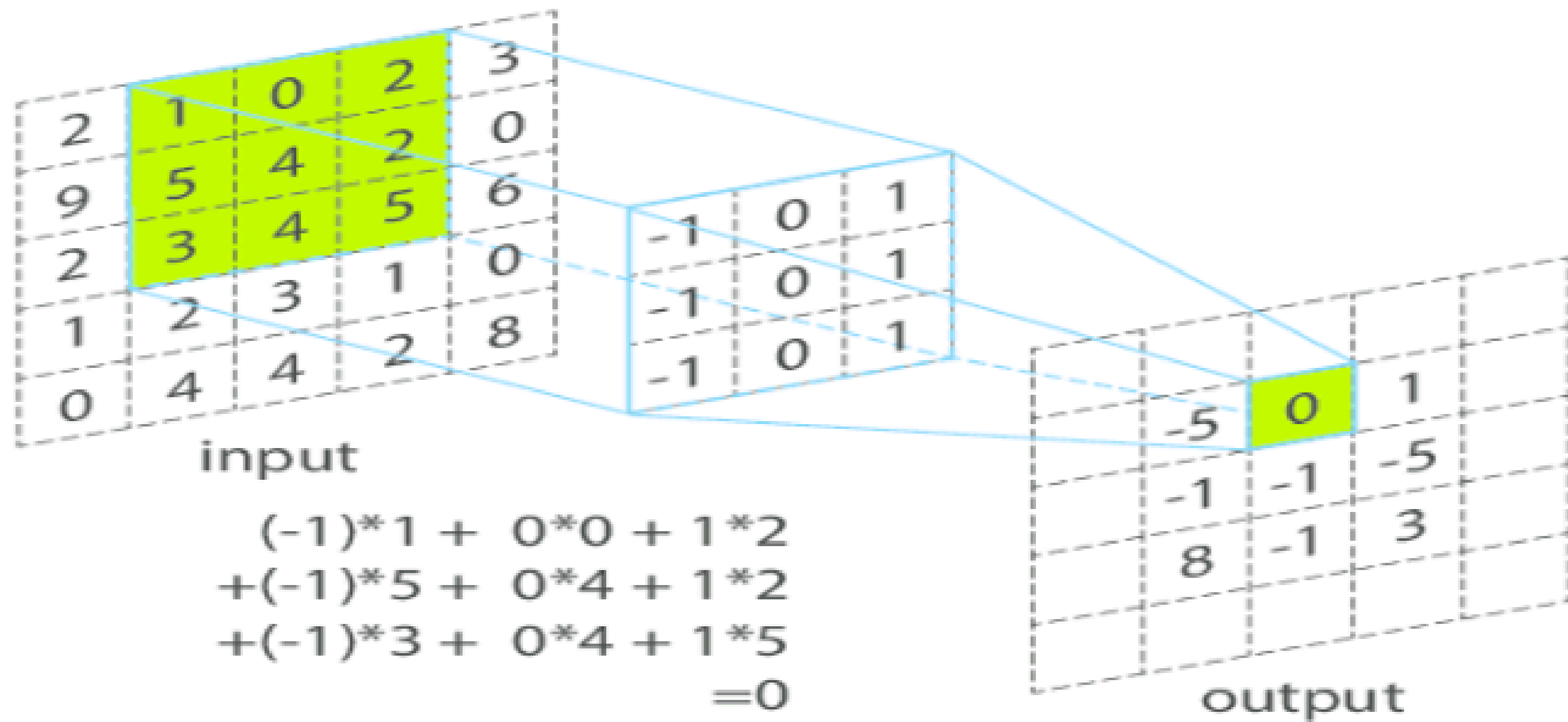
7 (7)



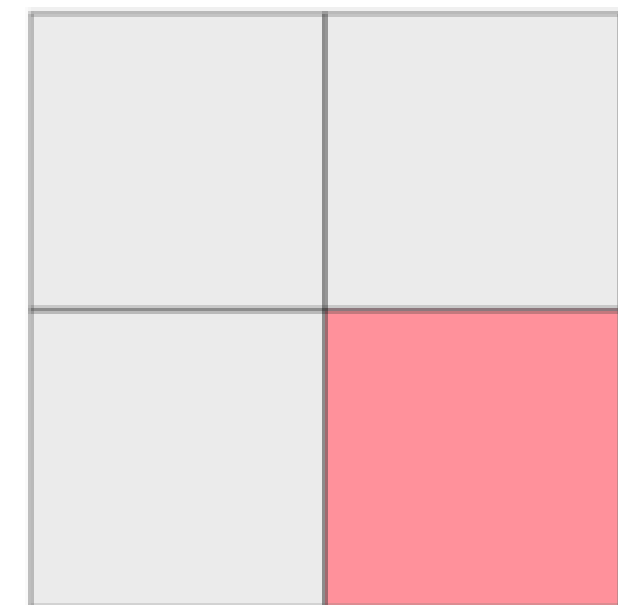
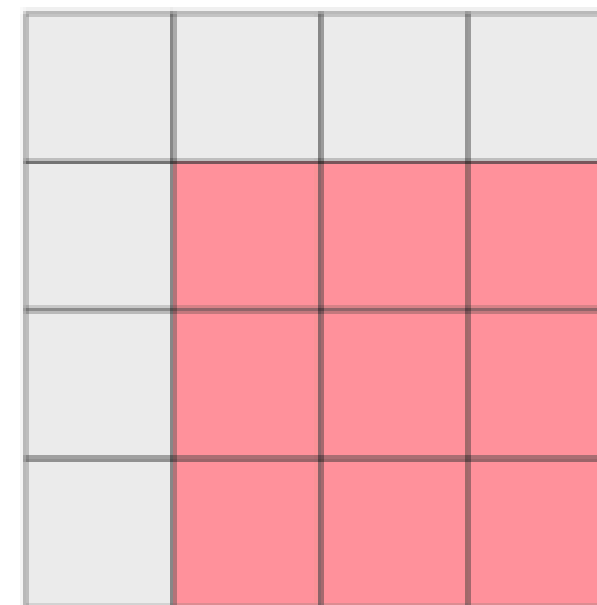
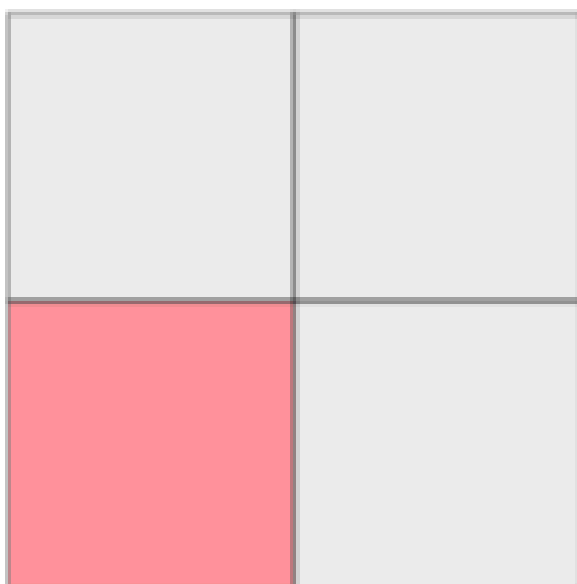
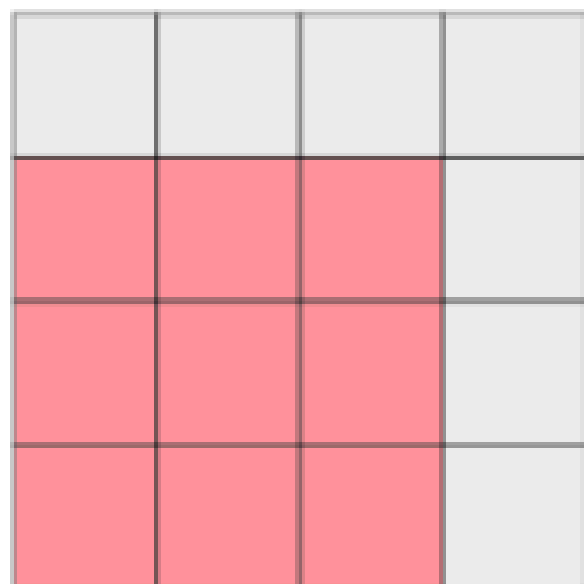
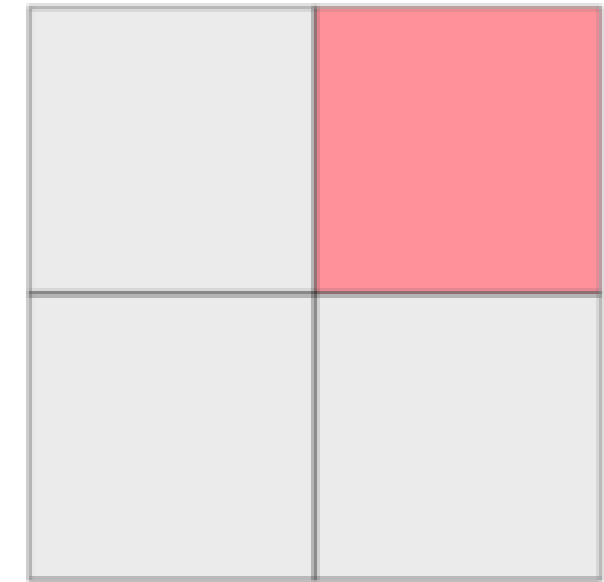
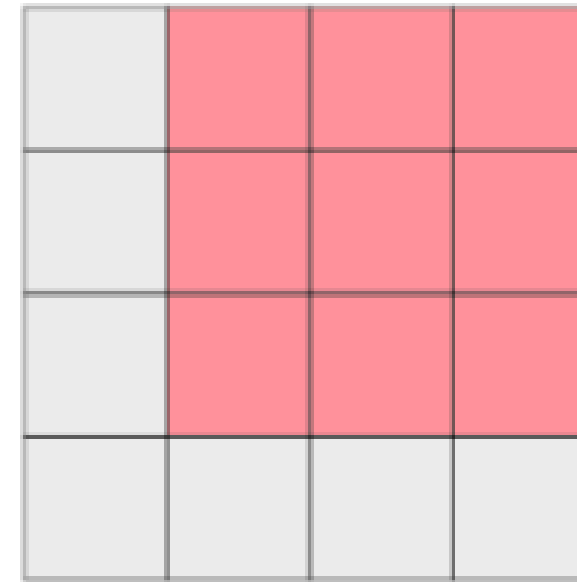
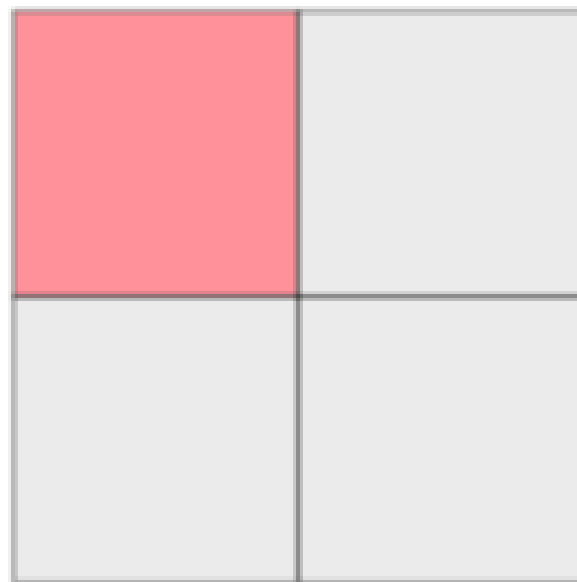
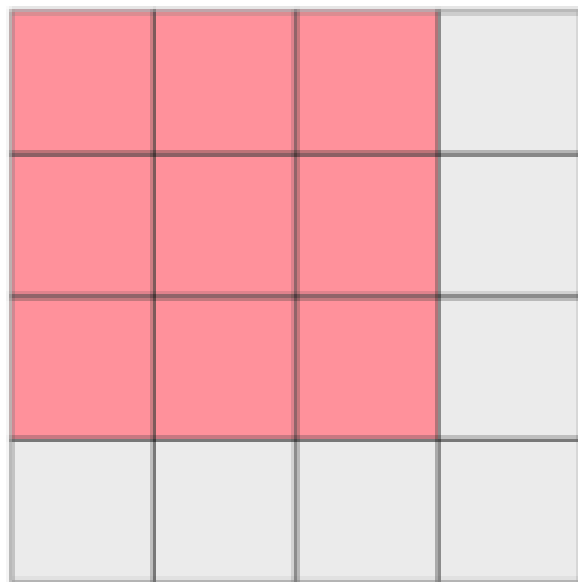
1 (1)

**reconnaitre certaines formes
permet de facilement conclure quel
chiffre est sur l'image !**

**ex : deux boucles → c'est un 8
une boucle → c'est un 8 un 6 ou un 9**

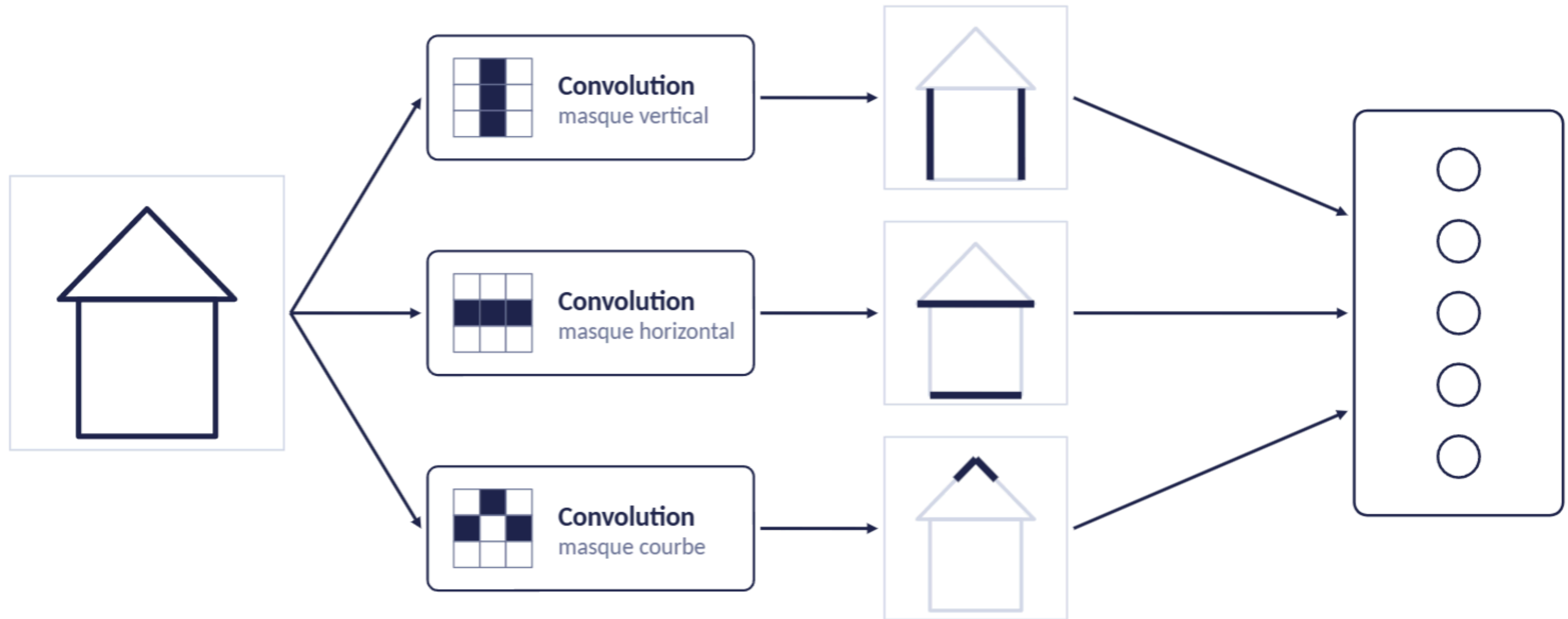


Les convolutions : formalisation



$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 3 & 0 \end{bmatrix}$$

**masque detecteur de colonne,
a une sortie "grande" si il y a une colonne a
cet emplacement**



au lieu de donner l'image au réseau, on lui donne la position de certains formes

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

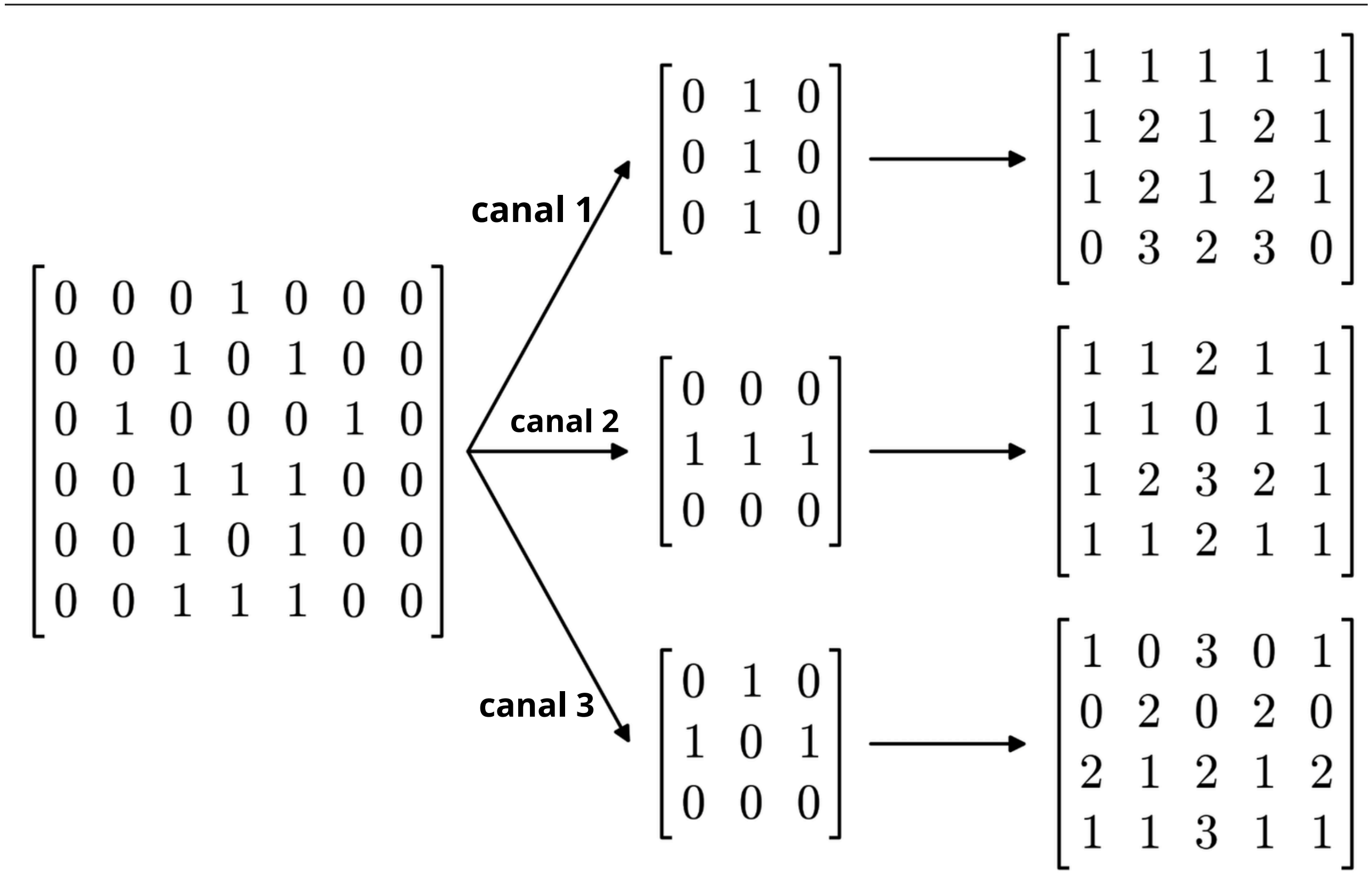
Trait vertical

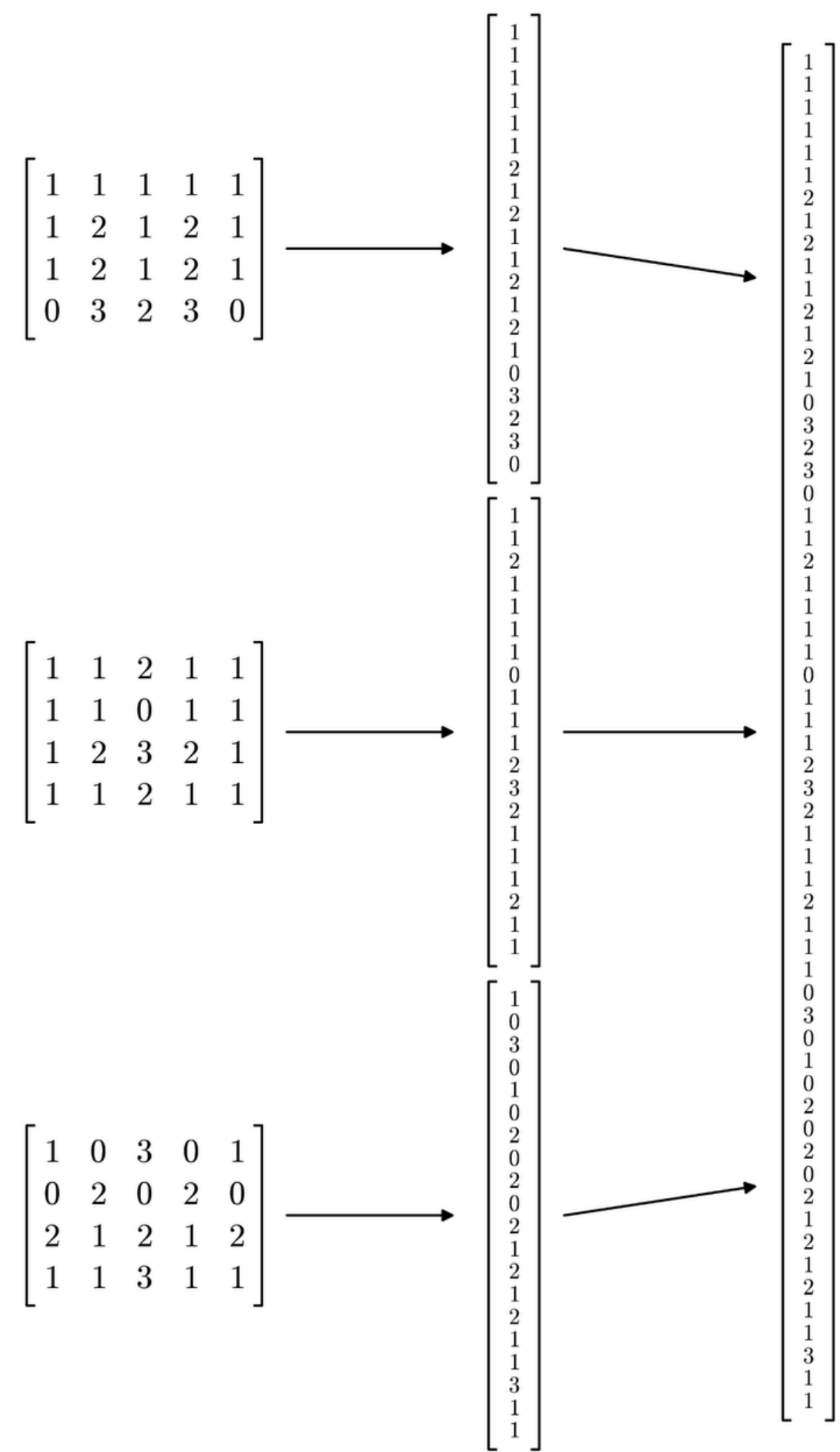
$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Trait horizontal

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Courbe





Comment on crée les masques ?

1. l est la l -ième couche, où $l = 1$ est la première couche et $l = L$ est la dernière couche.
2. L'entrée x est de dimension $H \times W$ et a i et j comme itérateurs.
3. Le filtre (ou noyau) w est de dimension $k_1 \times k_2$ et a m et n comme itérateurs.
4. $w_{m,n}^l$ est la matrice de poids reliant les neurones de la couche l aux neurones de la couche $l - 1$.
5. b^l est le biais de la couche l .

6. $x_{i,j}^l$ est le vecteur d'entrée convolué à la couche l , biais inclus, donné par :

$$x_{i,j}^l = \sum_m \sum_n w_{m,n}^l o_{i+m, j+n}^{l-1} + b^l = (w^l * o^{l-1})_{i,j} + b^l$$

7. $o_{i,j}^l$ est le vecteur de sortie à la couche l , donné par :

$$o_{i,j}^l = f(x_{i,j}^l)$$

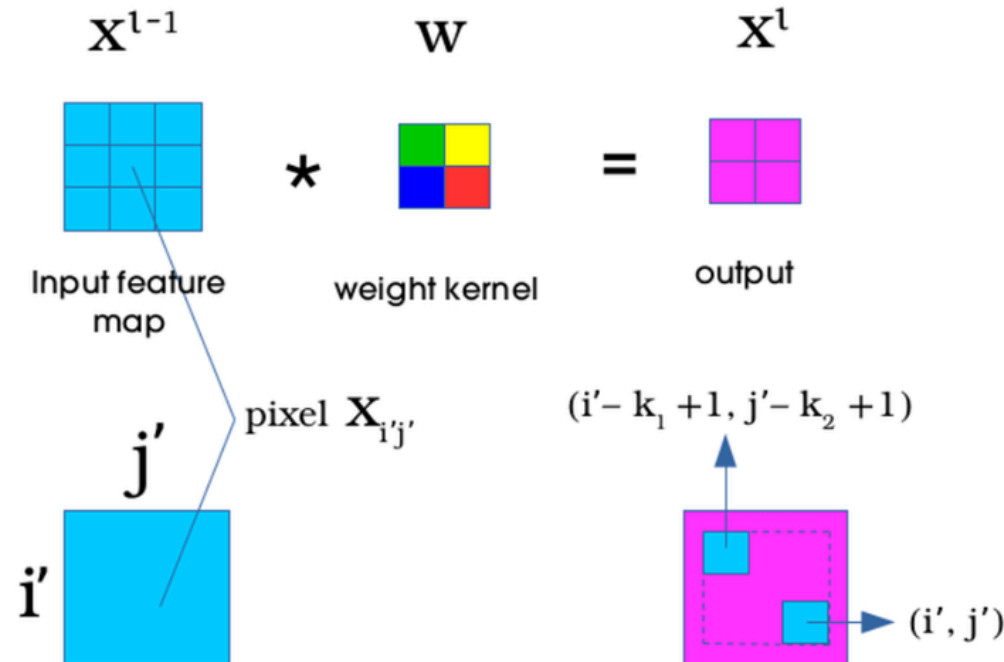
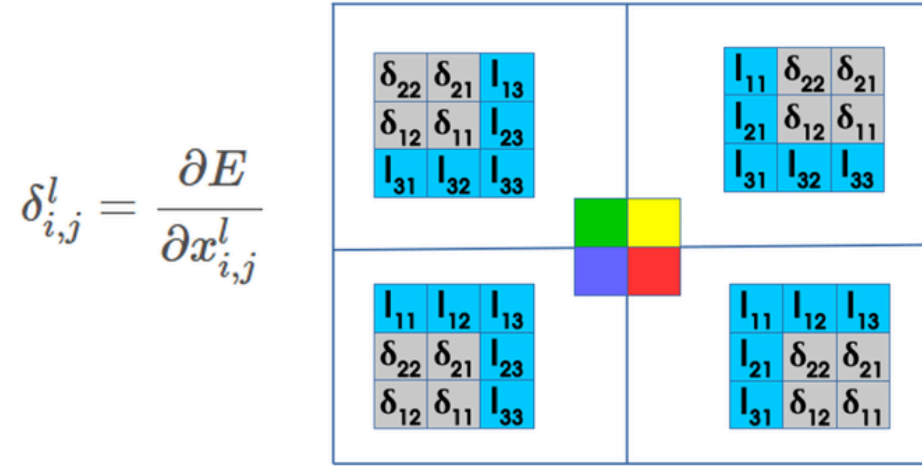
8. $f(\cdot)$ est la fonction d'activation. L'application de la couche d'activation au vecteur d'entrée convolué à la couche l est donnée par $f(x_{i,j}^l)$.

Note : $(w^l * o^{l-1})_{i,j}$ désigne le produit de convolution entre w^l et o^{l-1} , évalué en (i, j) — c'est exactement ce que représente la double somme ci-dessus.

$$\begin{aligned}\frac{\partial E}{\partial w_{m',n'}^l} &= \sum_{i=0}^{H-k_1} \sum_{j=0}^{W-k_2} \frac{\partial E}{\partial x_{i,j}^l} \frac{\partial x_{i,j}^l}{\partial w_{m',n'}^l} \\ &= \sum_{i=0}^{H-k_1} \sum_{j=0}^{W-k_2} \delta_{i,j}^l \frac{\partial x_{i,j}^l}{\partial w_{m',n'}^l}\end{aligned}$$

$$\frac{\partial x_{i,j}^l}{\partial w_{m',n'}^l} = \frac{\partial}{\partial w_{m',n'}^l} \left(\sum_m \sum_n w_{m,n}^l o_{i+m,j+n}^{l-1} + b^l \right) = \frac{\partial}{\partial w_{m',n'}^l} \left(w_{m',n'}^l o_{i+m',j+n'}^{l-1} \right) = o_{i+m',j+n'}^{l-1}$$

$$\frac{\partial E}{\partial w_{m',n'}^l} = \sum_{i=0}^{H-k_1} \sum_{j=0}^{W-k_2} \delta_{i,j}^l o_{i+m',j+n'}^{l-1}$$



$$\begin{aligned}\frac{\partial E}{\partial x_{i',j'}^l} &= \sum_{m=0}^{k_1-1} \sum_{n=0}^{k_2-1} \frac{\partial E}{\partial x_{i'-m,j'-n}^{l+1}} \frac{\partial x_{i'-m,j'-n}^{l+1}}{\partial x_{i',j'}^l} \\ &= \sum_{m=0}^{k_1-1} \sum_{n=0}^{k_2-1} \delta_{i'-m,j'-n}^{l+1} \frac{\partial x_{i'-m,j'-n}^{l+1}}{\partial x_{i',j'}^l}\end{aligned}$$

$$\begin{aligned}\frac{\partial x_{i'-m,j'-n}^{l+1}}{\partial x_{i',j'}^l} &= \frac{\partial}{\partial x_{i',j'}^l} \left(\sum_{m'} \sum_{n'} w_{m',n'}^{l+1} o_{i'-m+m',j'-n+n'}^l + b^{l+1} \right) \\ &= \frac{\partial}{\partial x_{i',j'}^l} \left(\sum_{m'} \sum_{n'} w_{m',n'}^{l+1} f \left(x_{i'-m+m',j'-n+n'}^l \right) + b^{l+1} \right)\end{aligned}$$

$$\begin{aligned}\frac{\partial x_{i'-m,j'-n}^{l+1}}{\partial x_{i',j'}^l} &= \frac{\partial}{\partial x_{i',j'}^l} \left(w_{m',n'}^{l+1} f \left(x_{0-m+m',0-n+n'}^l \right) + \dots + w_{m,n}^{l+1} f \left(x_{i',j'}^l \right) + \dots + b^{l+1} \right) \\ &= \frac{\partial}{\partial x_{i',j'}^l} \left(w_{m,n}^{l+1} f \left(x_{i',j'}^l \right) \right) \\ &= w_{m,n}^{l+1} \frac{\partial}{\partial x_{i',j'}^l} \left(f \left(x_{i',j'}^l \right) \right) \\ &= w_{m,n}^{l+1} f' \left(x_{i',j'}^l \right)\end{aligned}$$

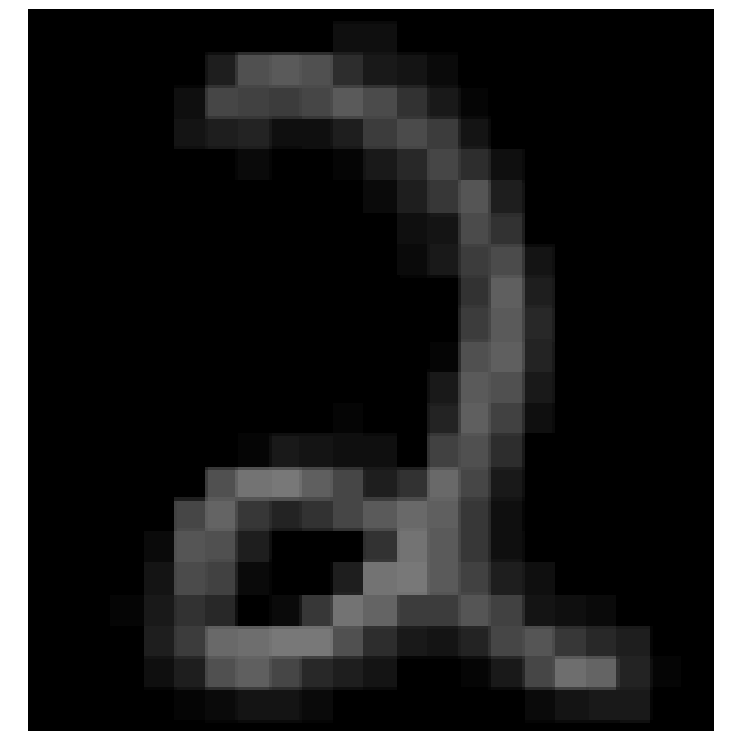
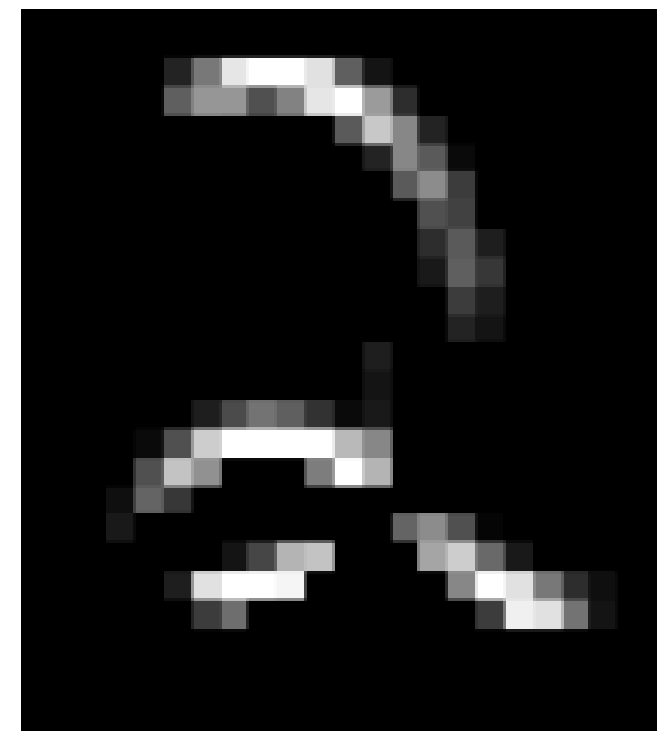
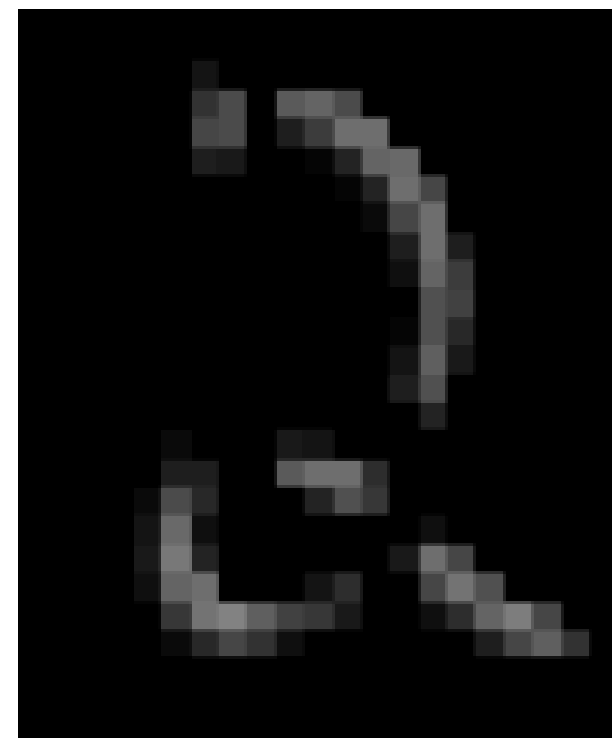
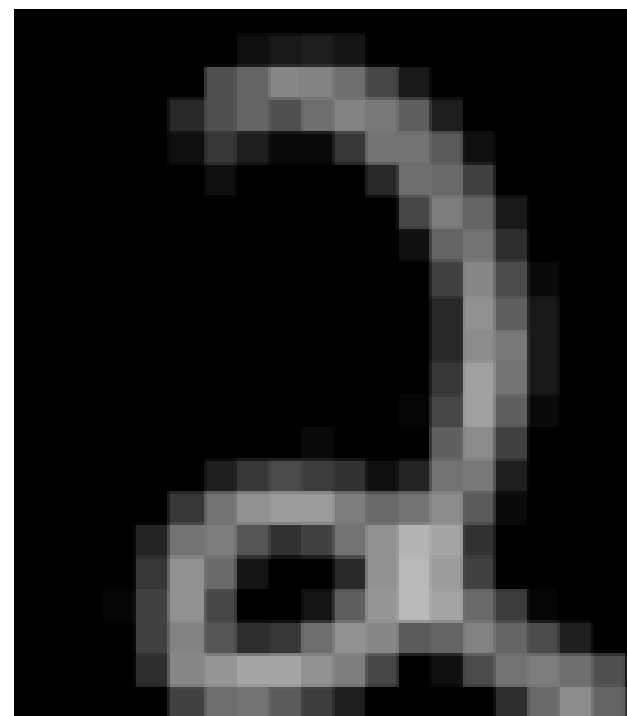
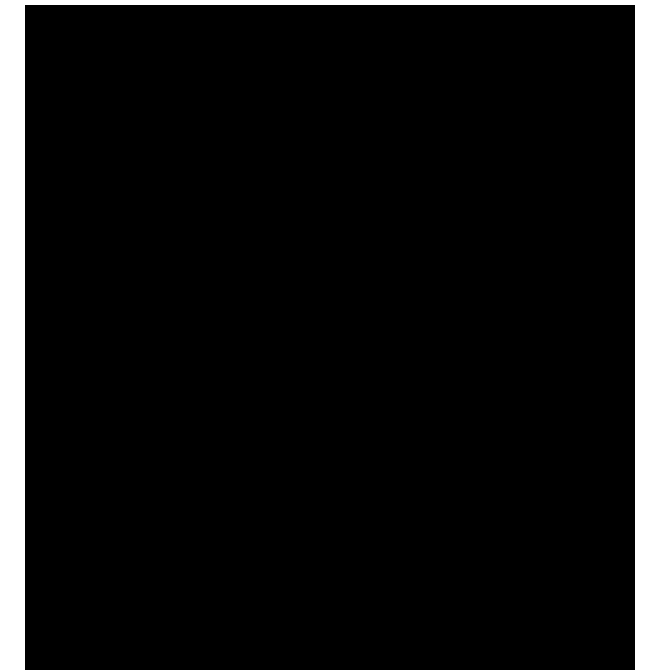
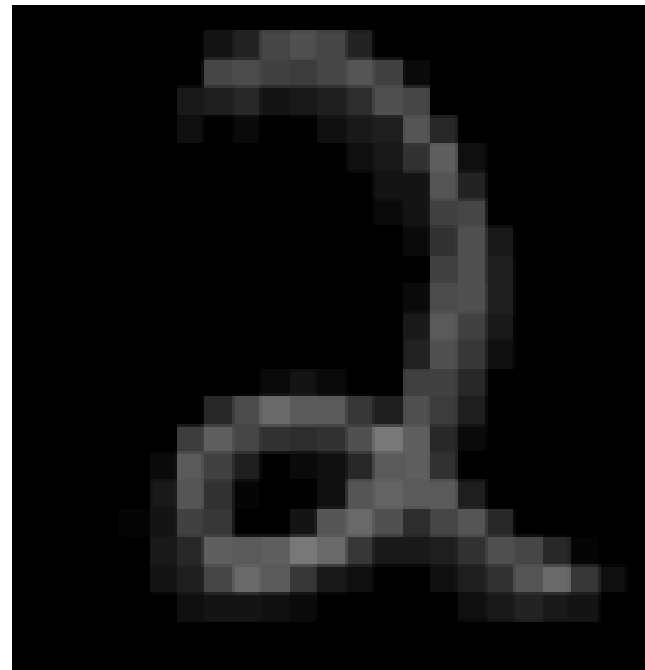
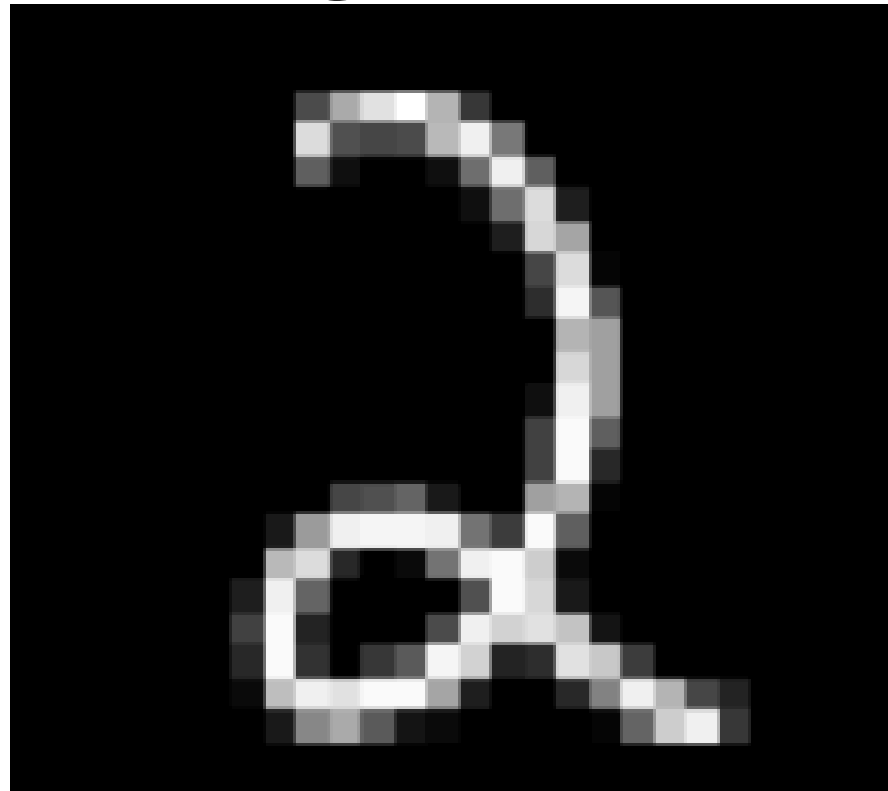
$$\frac{\partial E}{\partial x_{i',j'}^l} = \sum_{m=0}^{k_1-1} \sum_{n=0}^{k_2-1} \delta_{i'-m,j'-n}^{l+1} w_{m,n}^{l+1} f' \left(x_{i',j'}^l \right)$$

$$\frac{\partial E}{\partial w_{m',n'}^l} = \sum_{i=0}^{H-k_1} \sum_{j=0}^{W-k_2} \delta_{i,j}^l o_{i+m',j+n'}^{l-1} \quad w_{m',n'}^l \leftarrow w_{m',n'}^l - \eta \frac{\partial E}{\partial w_{m',n'}^l}$$

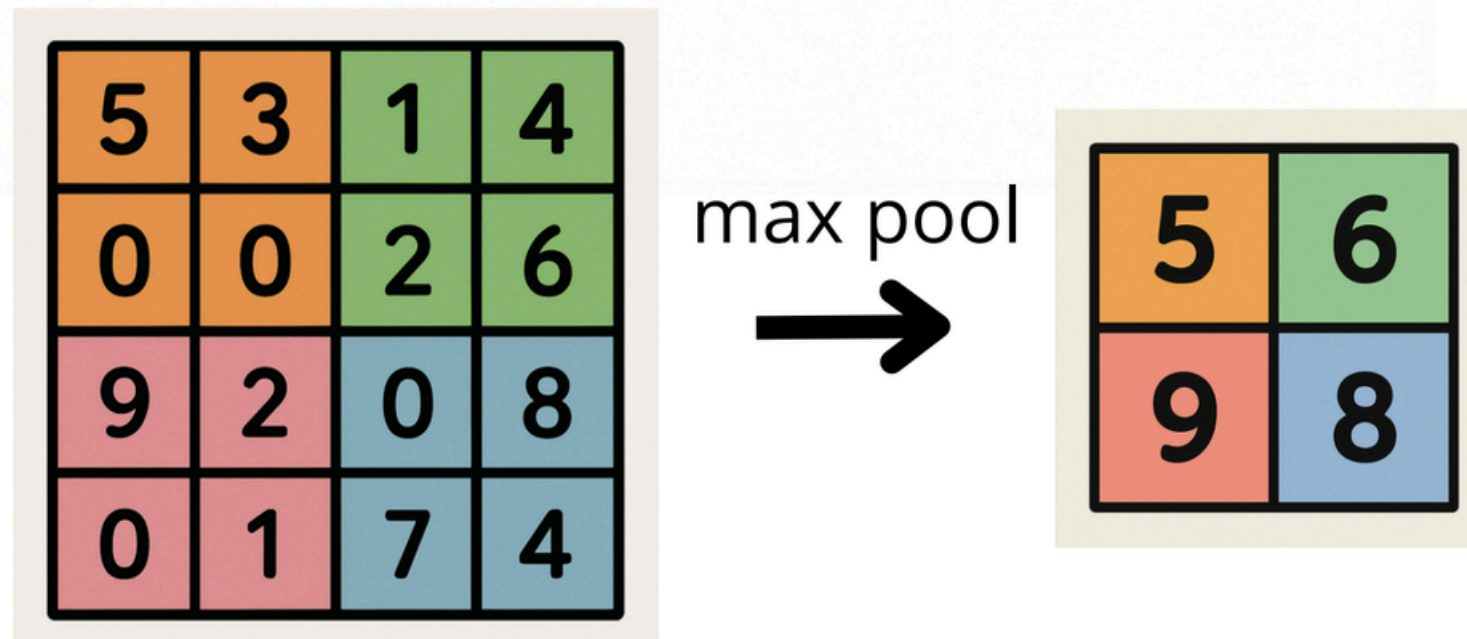
$$\frac{\partial E}{\partial b^l} = \sum_{i=0}^{H-k_1} \sum_{j=0}^{W-k_2} \delta_{i,j}^l \quad b^l \leftarrow b^l - \eta \frac{\partial E}{\partial b^l}$$

Les convolutions : exemple

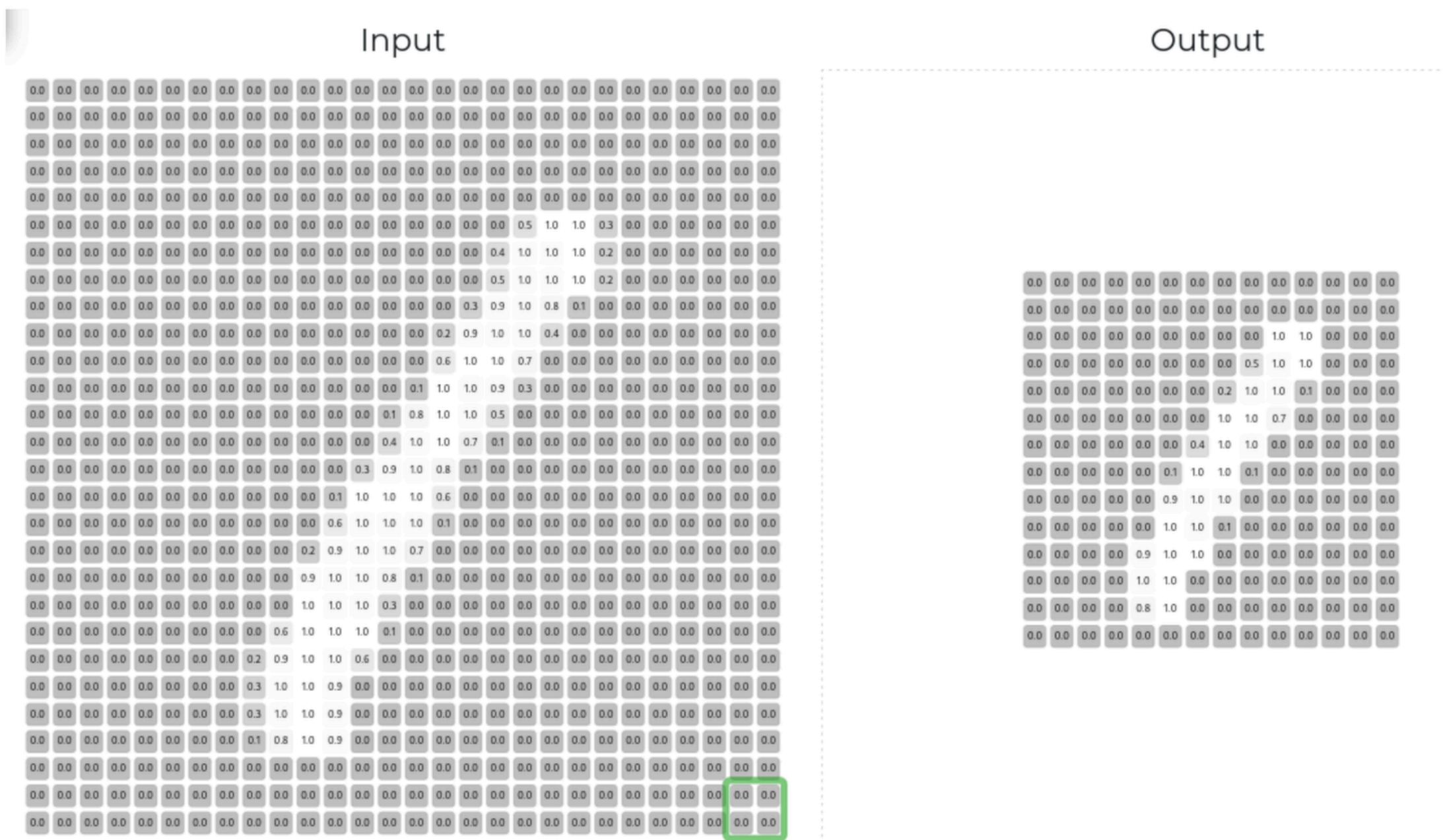
image de base



Max pooling :

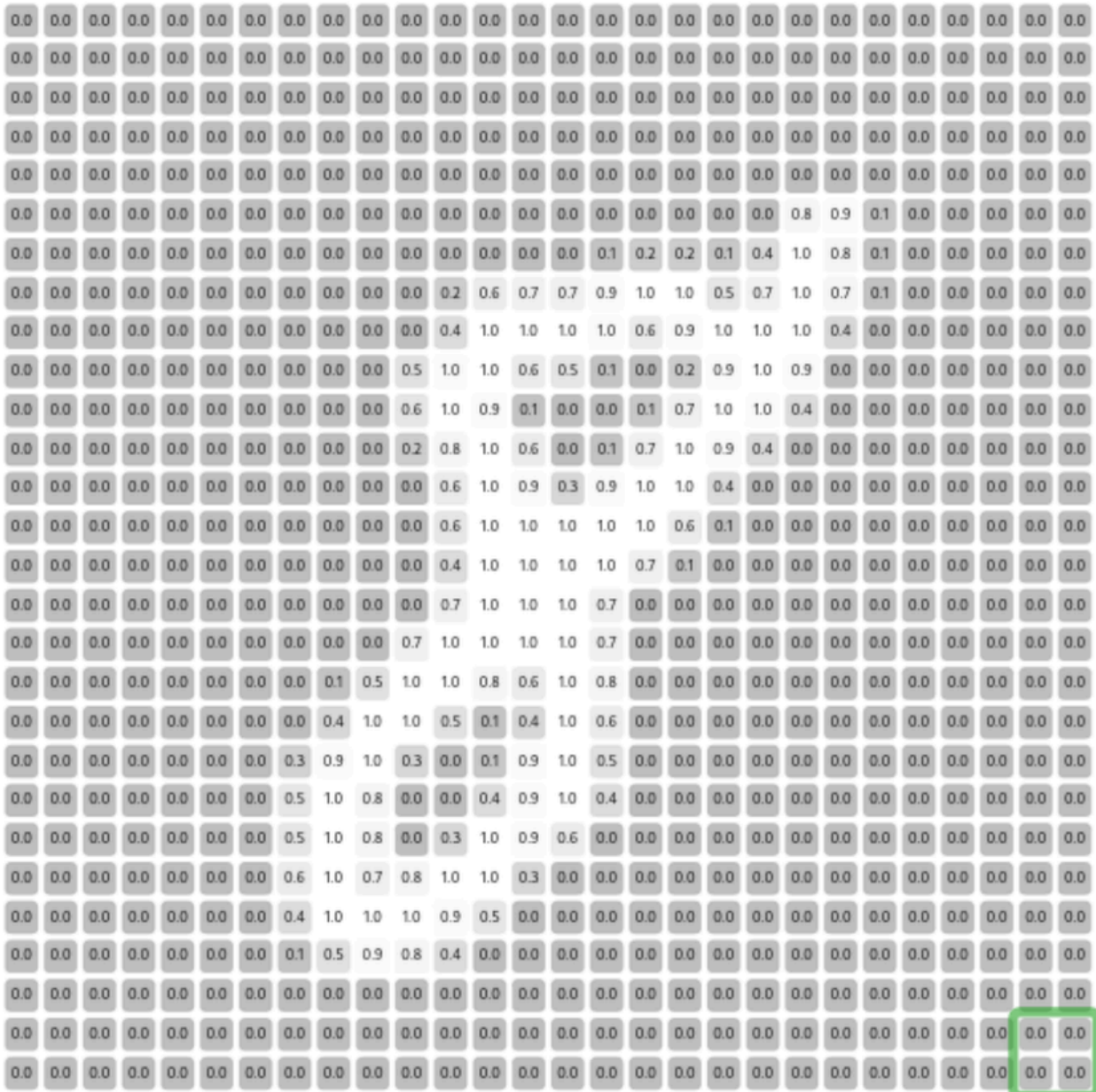


Effet du max pooling sur une image

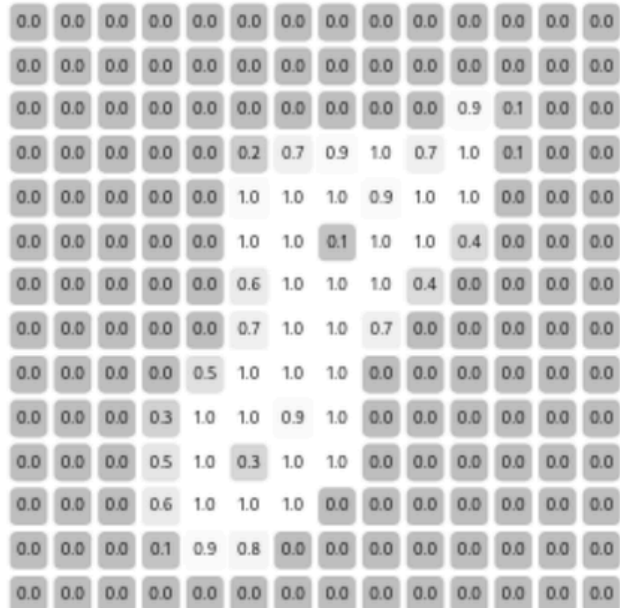


Effet du max pooling sur une image

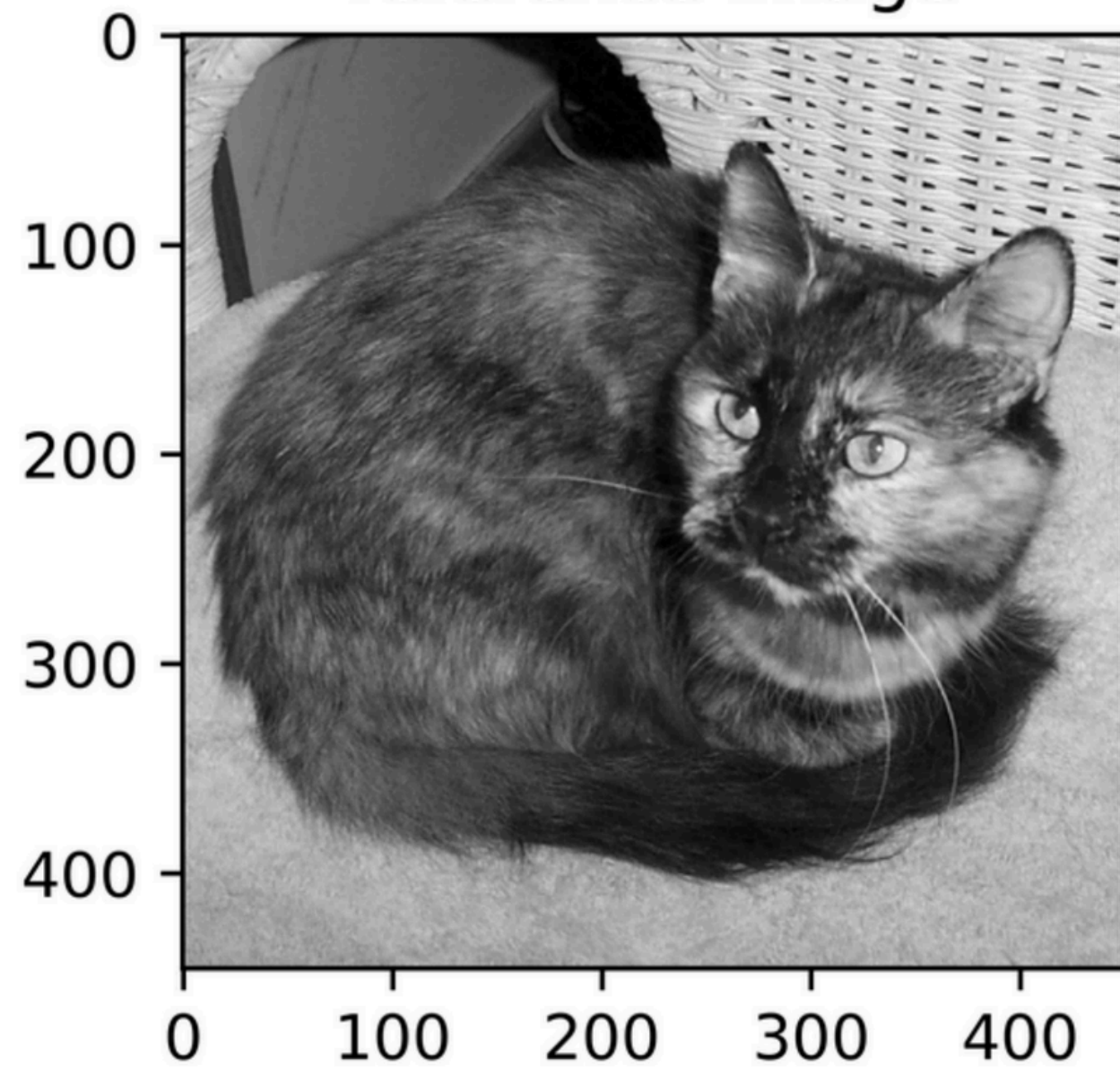
Input



Output



reference image



maxpooled

